ExpNo – 17 Experiment to understand Linear regression for a given set

Program:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
df.info()
df.dropna(inplace=True)
df.info()
df.describe()
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=1)
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
model.score(x_train,y_train)
model.score(x_test,y_test)
model.coef_
model.intercept_
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
model=pickle.load(open('SalaryPred.model','rb'))
```

```
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
print("Estimated Salary for {} years of experience is {}:
".format(yr_of_exp,Salary))
```

Output:

```
<class 'pandas.DataFrame'>
RangeIndex: 25 entries, 0 to 24
Data columns (total 2 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   YearsExperience  25 non-null    float64
 1   Salary          25 non-null    int64
dtypes: float64(1), int64(1)
memory usage: 532.0 bytes
<class 'pandas.DataFrame'>
RangeIndex: 25 entries, 0 to 24
Data columns (total 2 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   YearsExperience  25 non-null    float64
 1   Salary          25 non-null    int64
dtypes: float64(1), int64(1)
memory usage: 532.0 bytes
Enter Years of Experience:  6
```

Estimated Salary for 6.0 years of experience is [[83465.18762507]]: